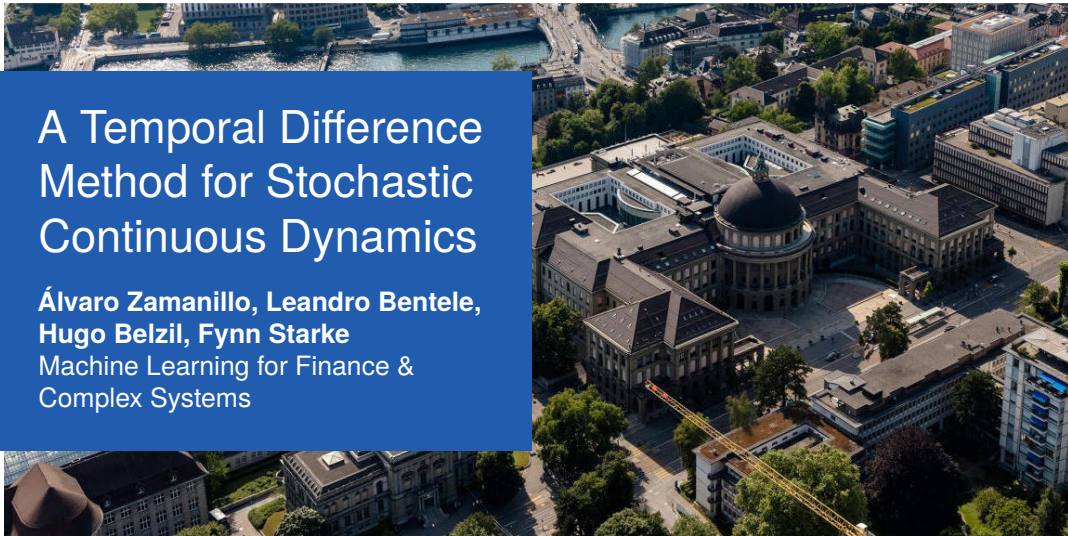


A Temporal Difference Method for Stochastic Continuous Dynamics

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Outline

1. TD Learning

2. dTD learning

3. Implementing dTD for the Merton Problem

Reinforcement learning in a nutshell

An agent in states $s \in \mathcal{S}$ takes actions $a \sim \pi(\cdot \mid s)$ and observes rewards $r(s, a)$.

Goal: maximise the expected discounted return

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{k=0}^{\infty} \gamma^k r(s_k, a_k) \mid s_0 = s \right].$$

Policy evaluation: learn V^π from sampled transitions (s, a, s', r) .

The Bellman equation and TD learning

Bellman equation (fixed policy π)

$$V^\pi(s) = \mathbb{E}_{s' \sim P^\pi(\cdot|s)} [r(s, s') + \gamma V^\pi(s')].$$

TD error and value loss

$$\delta_t = \underbrace{r_t + \gamma V_\theta(s_{t+1})}_{\text{target (detached)}} - \underbrace{V_\theta(s_t)}_{\text{prediction}}, \quad L_V(\theta) = \frac{1}{2} \mathbb{E}[\delta_t^2].$$

This is a simple online model-free method!

Actor-critic

Two networks working together:

- *Actor* — the policy: proposes actions given the current state.
- *Critic* — the value function: scores how good states are.

The TD error tells the critic how wrong its prediction was, and tells the actor whether the action it just took was better or worse than expected.

- If TD error > 0 : the action beat expectation $\Rightarrow$ make it more likely.
- If TD error < 0 : the action underperformed $\Rightarrow$ make it less likely.

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Continuous-time RL

Continuous-time RL

State space $\mathcal{S} \subset \mathbb{R}^n$, action space $\mathcal{A}$, controlled SDE

$$dS_t = \mu(S_t, A_t) dt + \sigma(S_t, A_t) dB_t, \quad A_t \sim \pi(\cdot \mid S_t).$$

$$\mu : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^n$$

$$\sigma : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^{n \times m}$$

$(B_t)_{t \geq 0}$ is the m -dimensional Brownian motion.

- Many physical and financial systems are naturally continuous (robotics, autonomous driving, portfolio optimisation, ...).
- **Model-free assumption:** The agent has no prior knowledge of (or incentive to learn) μ or σ and observes the system only at discrete instants $t, t + \Delta t, t + 2\Delta t, \dots$

The Hamilton–Jacobi–Bellman (HJB) equation

Bellman equation after discretisation

$$V^\pi(s_t) = \mathbb{E}_{p^\pi} \left[\rho(s_t, A_t) \Delta t + e^{-\gamma \Delta t} V^\pi(S_{t+\Delta t}) \right],$$

HJB equation (obtained by Taylor Expansion and Itô on $V^\pi(S_{t+\Delta t})$)

$$V^\pi(s) = \frac{1}{\gamma} \mathbb{E}_{p^\pi} \left[\rho(s, A) + \mu(s, A) \cdot \nabla V^\pi(s) + \frac{1}{2} \text{tr}(\sigma \sigma^\top(s, A) \nabla^2 V^\pi(s)) \right].$$

- **Bellman:** continuity hidden in the *transition kernel* P^π — sampling destroys it.
- **HJB:** continuity *inside the expectation*, via $\mu, \sigma \sigma^\top, \nabla V, \nabla^2 V$ — it survives sampling.
- *Catch:* the update needs μ, σ

Drift and diffusion from samples

Proposition (Settai, Takeishi, Yairi 2025)

For the SDE $dS_t = \mu dt + \sigma dB_t$,

$$\mathbb{E}_{p^\pi}[\mu^i(s, A)] = \lim_{\Delta t \rightarrow 0} \mathbb{E}_{p^\pi} \left[\frac{S_{t+\Delta t}^i - s_t^i}{\Delta t} \right],$$

$$\mathbb{E}_{p^\pi}[(\sigma\sigma^\top)^{ij}] = \lim_{\Delta t \rightarrow 0} \mathbb{E}_{p^\pi} \left[\frac{(S_{t+\Delta t}^i - s_t^i)(S_{t+\Delta t}^j - s_t^j)}{\Delta t} \right].$$

$\Rightarrow$ We can replace $\mu, \sigma\sigma^\top$ by sampled increments of S_t .

Substituting the sample estimators into the HJB equation gives a model-free residual.

dTD residual

$$\text{dTD} = \frac{1}{\gamma} \left(\rho(s_t, A_t) + \frac{\Delta s_t}{\Delta t} \cdot \nabla V(s_t) + \frac{1}{2} \frac{1}{\Delta t} \Delta s_t^\top \nabla^2 V(s_t) \Delta s_t \right) - V(s_t),$$

with $\Delta s_t = s_{t+\Delta t} - s_t$.

TD vs. naive-dTD vs. dTD

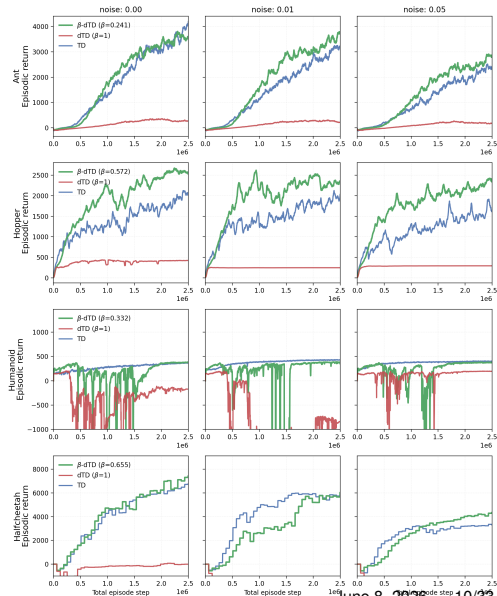
Method	Target (fixed)	Prediction (detached)
TD	$r(s_t, a_t) + \gamma_{\text{discrete}} V(s_{t+1})$	$V(s_t)$
naive-dTD	$\rho(s_t, a_t) + \frac{\Delta s_t}{\Delta t} \cdot \nabla V + \frac{1}{2\Delta t} \Delta s_t^\top \nabla^2 V \Delta s_t$	$\gamma V(s_t)$
dTD	$-\rho(s_t, a_t) + \gamma V(s_t)$	$\frac{\Delta s_t}{\Delta t} \cdot \nabla V + \frac{1}{2\Delta t} \Delta s_t^\top \nabla^2 V \Delta s_t$

- Target is kept constant, the prediction is used in gradient descent
- In practice, function approximation makes simple dTD unstable. Define **β -dTD**:

$$\mathcal{L}_{\beta\text{-dTD}} = (1 - \beta) \mathcal{L}_{\text{TD}} + \beta \mathcal{L}_{\text{dTD}}$$

Reported experiments

- Environments: Hopper, HalfCheetah, Ant, Humanoid (dim. 11–244).
- Not an SDE: noise only in the observation, $\tilde{s}^i \leftarrow s^i + c |s^i| \varepsilon$, with $c \in \{0, 0.01, 0.05\}$.
- β -dTD is comparable to or better than TD across the tested settings.
- Tuned β for β -dTD is non-negligible.
- For β -naive-dTD, the tuned β is close to 0, i.e. near the standard TD update (not plotted)



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Merton Problem

Wealth dynamics:

$$\frac{dW_t}{W_t} = (r + \pi_t(\mu_{BS} - r) - \kappa_t) dt + \pi_t \sigma_{BS} dB_t, \quad c_t = \kappa_t W_t.$$

Parameters: risk-free rate r , expected return μ_{BS} , volatility σ_{BS} , risky-asset fraction π_t , consumption rate κ_t .

Objective:

$$\mathbb{E} \left[\int_0^\infty e^{-\rho t} U(c_t) dt \right], \quad U(c) = \frac{c^{1-\gamma}}{1-\gamma}, \quad 0 < \gamma < 1.$$

- Falls in the setup of the paper: stochastic continuous dynamics.
- **Notation note:** here ρ is the subjective discount rate and γ the CRRA coefficient.

Analytical solution of the Merton problem

- Optimal policy: fraction of wealth invested in risky asset and fraction of wealth consumed are constant:

$$\pi^* = \frac{\mu_{BS} - r}{\gamma \sigma_{BS}^2}, \quad \kappa^* = \frac{\rho - (1 - \gamma) \left(r + \frac{1}{2} \frac{(\mu_{BS} - r)^2}{\gamma \sigma_{BS}^2} \right)}{\gamma}.$$

Value function for the Merton problem

- For the optimal policy the value function is:

$$V^{\pi, \kappa}(w) = A(\pi, \kappa) \frac{w^{1-\gamma}}{1-\gamma},$$

$$A(\pi, \kappa) = \frac{\kappa^{1-\gamma}}{\rho - (1-\gamma) \left[r + \pi(\mu_{BS} - r) - \kappa - \frac{1}{2} \gamma \pi^2 \sigma_{BS}^2 \right]}.$$

Training

Fix optimal policy and try to learn V^{π^*, κ^*}

Algorithm 1 Infinite-Horizon dTD Learning

- 1: **Initialize** critic parameters θ , and initial states w_0
 - 2: **for** each iteration **do**
 - 3: Sample transitions $(w_t, w_{t+\Delta t}, r_t)$ under fixed policy (π^*, κ^*)
 - 4: Compute dTD loss $\mathcal{L}(\theta)$
 - 5: Update θ via gradient descent on $\mathcal{L}$
 - 6: $w_t \leftarrow w_{t+\Delta t}$
 - 7: **end for**
-

Results for Pure dTD

Results: dTD collapses to a constant

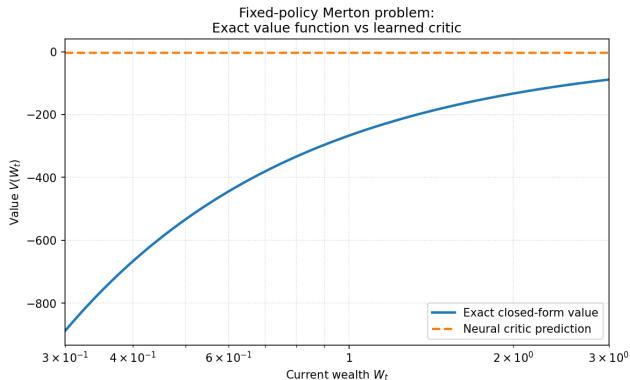


Figure: Value function learned with pure dTD loss

Results: dTD on its own does not learn the value function

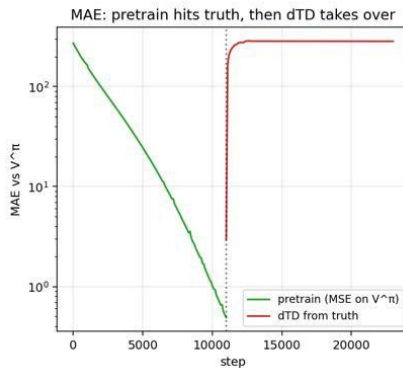


Figure: MAE for learned value function and its derivative starting from a pre-trained network

dTD: the prediction is noise-dominated

dTD prediction

$$\text{pred} = \frac{\Delta s}{\Delta t} \nabla V(s_t) + \frac{1}{2} \frac{(\Delta s)^2}{\Delta t} \nabla^2 V(s_t)$$

Decomposing the first term via Itô

Since $\Delta s \approx \mu \Delta t + \sigma \Delta B$:

$$\frac{\Delta s}{\Delta t} \nabla V = \underbrace{\mu \nabla V}_{\text{signal}} + \underbrace{\sigma \frac{\Delta B}{\Delta t} \nabla V}_{\text{noise, std} \sim \|\sigma \nabla V\| / \sqrt{\Delta t}}$$

As $\Delta t \rightarrow 0$, the noise term dominates. Spurious attractor: $\nabla V \rightarrow 0$.

Results for β -dTD

$$\mathcal{L} = (1 - \beta) \mathcal{L}_{TD} + \beta \mathcal{L}_{dTD}$$

Results β -dTD: The more noise in the SDE, the more TD component we need

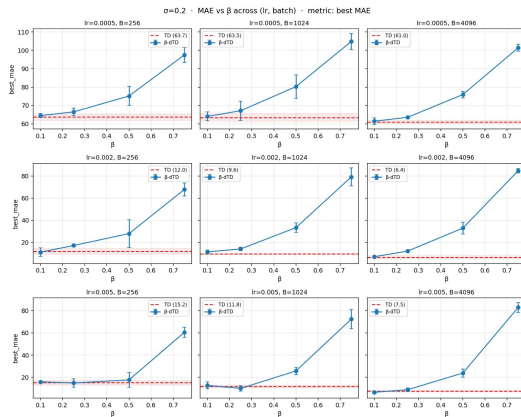


Figure: Hyperparameter performance for β -dTD with **high** noise

Results β -dTD: The more noise in the SDE, the more TD component we need

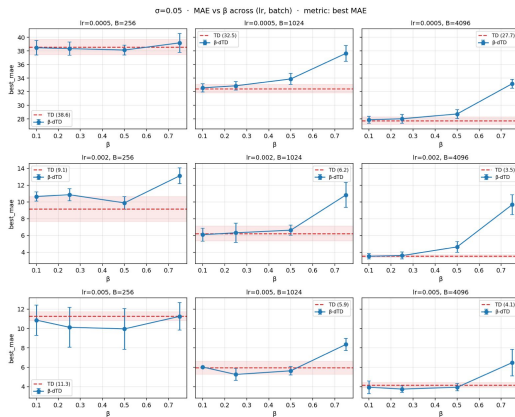


Figure: Hyperparameter performance for β -dTD with **low** noise

Results β -dTD: Cannot reproduce the "faster rate of improvement of β -dTD over TD"

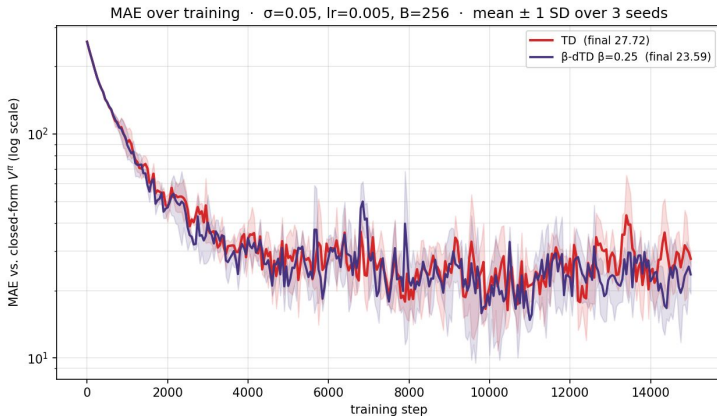


Figure: Evolution of MAE over training steps

Variance reduction for the drift estimator

Mean-target experiment: variance reduction with K -samples

Main idea

For each sampled wealth $w_{i,t}$, instead of drawing one transition, draw K independent next states:

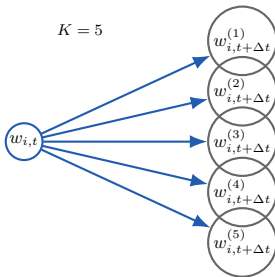
$$w_{i,t+\Delta t}^{(1)}, \dots, w_{i,t+\Delta t}^{(K)} \quad \text{from the same } w_{i,t}.$$

Then average the residual over the K transitions before squaring.

$K = 1$



$K = 5$



$$\Rightarrow \frac{1}{K} \sum_{k=1}^K \delta_{\beta\text{-dTD}}^{(k)}$$

Effect of K -samples on pure dTD

By using K -samples we do no longer get a constant learned value function; however, the computations needed scale by a factor of K .

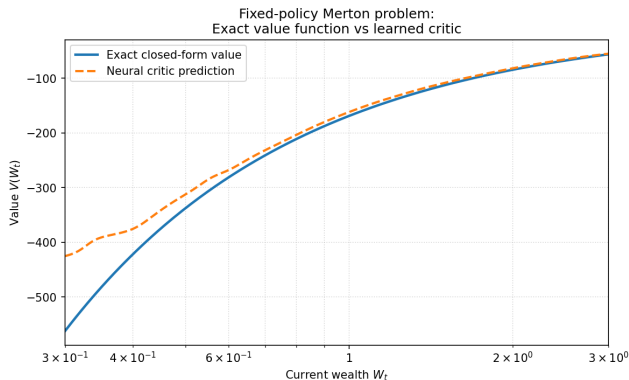
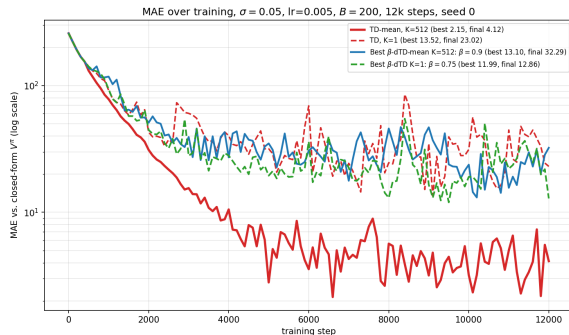


Figure: Value function learned with pure dTD loss and $K=10,000$

K -samples experiments: results

Method	K	Best MAE averaged over seeds
TD	1	13.52
β -dTD ($\beta = 0.75$)	1	13.10
TD	512	2.15
β -dTD ($\beta = 0.9$)	512	11.99



- higher K transitions reduces sampling noise.
- β -dTD does not substantially profit from higher K , while TD does

Merton problem with fixed horizon

Same problem as before, except we stop the problem at time $T < \infty$. For a fixed policy (π, κ) , with $C_s = \kappa W_s$, we evaluate

$$V^{\pi, \kappa}(t, w) = \mathbb{E} \left[\int_t^T e^{-\rho(s-t)} u(\kappa W_s) ds + e^{-\rho(T-t)} A_T u(W_T) \middle| W_t = w \right], \quad (1)$$

subject to the same wealth dynamics.

Under constant policies (π, κ) , the closed-form solution is

$$V^{\pi, \kappa}(t, w) = \underbrace{\left[A_T e^{-D(T-t)} + \frac{\kappa^{1-\gamma}}{D} (1 - e^{-D(T-t)}) \right]}_{:=A(t)} \frac{w^{1-\gamma}}{1-\gamma},$$

where

$$D = \rho - (1 - \gamma) \left[r + \pi(\mu - r) - \kappa - \frac{1}{2} \gamma \pi^2 \sigma^2 \right].$$

Finite-horizon HPO setup

Hyperparameter sweep ($K = 1024$)

$$\begin{aligned}\sigma &\in \{0.05, 0.20\}, & lr &\in \{5 \cdot 10^{-4}, 2 \cdot 10^{-3}, 5 \cdot 10^{-3}\}, \\ B &\in \{256, 1024, 4096\}, & \beta &\in \{0.1, 0.25, 0.5, 0.75\}.\end{aligned}$$

- MLP $V(w)$ becomes $V(w, \frac{t}{T})$ to account for time dependence

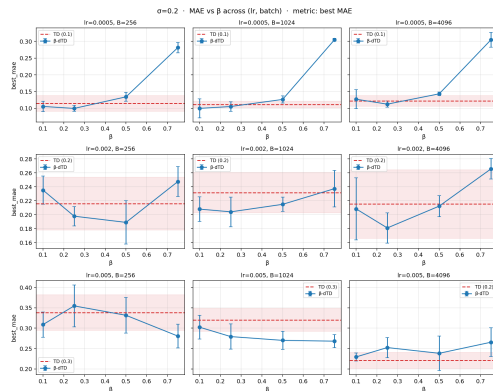
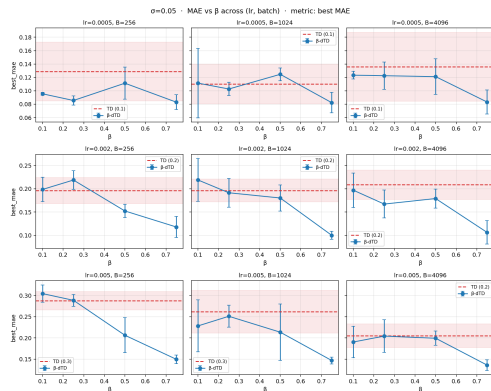
Finite-horizon HPO: takeaway

Noise level	Best TD MAE	Best β -dTD MAE	Best β
$\sigma = 0.05$	0.11	0.08	0.75
$\sigma = 0.20$	0.11	0.10	0.10–0.25

- For $\sigma = 0.05$, β -dTD improves over TD in all 9/9 tested (lr, B) combinations.
- For $\sigma = 0.20$, β -dTD improves over TD in 8/9 tested (lr, B) combinations.
- Low-noise case: the best β is consistently high, $\beta = 0.75$.
- Higher-noise case: the best β is smaller and more sensitive to (lr, B) .

Hypothesis: FH makes β -dTD look more useful than in the previous infinite-horizon experiments, but high diffusion still makes the dTD component less effective.

Finite-horizon HPO: MAE versus β



- At $\sigma = 0.05$, the curve favors large β .
- At $\sigma = 0.20$, the optimal β shifts lower, suggesting stronger TD regularization is needed.
- **Drawback** : K -samples are unrealistic?

Conclusion

- dTD on its own completely fails to learn the value function, we need TD to anchor the convergence.
- The method is very sensitive to process noise.
- K-samples reduce the variance in the drift estimator; however, it is computationally more expensive.
- On finite horizon, β -dTD performs better (theory: due to better problem regularization).

Possible Extension Work

- Implement multi-asset Merton problem and test if a higher dimension of the state space makes dTD more useful.
- Use Actor-Critic instead of fixed policy.
- Experiment with other techniques to reduce the variance of the drift estimator
- Dynamic β : exploit the possible benefits of dTD at the beginning of training, switch to TD later on.

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June 8, 2026